GG-A-440C
November 18, 1981
SUPERSEDING
GG-A-440B
October 13, 1976

FEDERAL SPECIFICATION

ALIDADES, SURVEYING: TELESCOPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers telescopic surveying alidades utilized in accomplishing topographic and reconnaissance surveys by the planetable method.

1.2 Classification.

1.2.1 Types. Surveying alidades shall be of the following types, as specified (see 6.2):

Type III - Optical-reading, self-leveling, high standard.
Type IV - Optical-reading, self-leveling, low standard.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6675

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Federal Standards:

Fed. Std. No. H28 - Screw-Thread Standards for Federal Services.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal Specifications, and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Description from established distribution points in their agencies.)

Military Specifications:

MIL-C-675 - Coating of Glass Optical Elements (Anti-Reflection).
MIL-S-3534 - Surveying Instruments and Accessories, Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-170 - Moisture Resistance Test Cycle for Ground Signal and Electronic Test Equipment.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

ASTM B 36 - Brass Plate, Sheet, Strip, and Rolled Bar.
ASTM B 209 - Aluminum-Alloy Sheet and Plate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Department of Commerce:

NBS Special Publication 374 - Methods for Determining the Resolving Power of Photographic Lenses

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Society of Automotive Engineers, Inc. (SAE):

HS 1086 - Unified Numbering System for Metals and Alloys.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

3. REQUIREMENTS

3.1 Description. The alidades shall consist of a blade upon which a circular level for leveling the planetable, a compass, and standards, or a pedestal with standards supporting a telescope, are mounted. The telescope shall be equipped with cross and stadia hairs and a vertical motion clamp or clutch with tangent screw.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Unified numbering system (UNS). Where applicable, metal alloys are identified by UNS in accordance with SAE HS 1086.

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3.4 Screw threads. All screw threads shall conform to Fed. Std. No. H28.

3.5 Environmental requirements.

3.5.1 Operating temperature. The alidade shall perform as specified herein in any ambient temperature from minus 25° F to plus 110° F.

3.5.2 Storage temperature. The alidade shall not be damaged when stored under the following conditions:

(a) High - Plus 145° F for 4 consecutive hours.
(b) Low - Minus 28° F for 6 consecutive hours.

3.5.3 Solar radiation. The alidade shall not be damaged by solar radiation of 355 British thermal units per square foot per hour (Btu/sq ft/hr) for a period of not less than 4 hours at an air temperature of plus 110° F.

3.5.4 Operating humidity. The alidade shall not be damaged when subjected to one complete humidity cycle in accordance with MIL-STD-170.

3.6 Optical system. The optical system shall produce a sharp image throughout the entire focusing range. All internal lens-air surfaces, except the registered face of the reticle, shall be coated in accordance with MIL-C-675. The field of view shall be substantially flat from the center to a zone outside the stadia at least one-fourth the stadia interval and shall show no marked curvature to the edge of the field. The entire field of view shall subtend an angle of not less than 1 degree 30 minutes from bottom to top. The optical elements shall be mounted so that the optical center of the focusing lens shall move parallel to the line of collimation.

3.6.1 Angle of resolution. The limiting angle of resolution shall be 5 seconds, plus or minus 1 second of arc.

3.7 Telescope. The telescope shall be the inverting, internal-focusing, dustproof, and moistureproof type. The system shall be set up so that the line of collimation is parallel to the axis of the telescope. The precision of the focusing movements shall be such that the direction of the line of collimation shall not change more than 5 seconds for the complete focusing range. The focusing ring or knob shall be located near the eye end of the telescope. The minimum focus shall be not more than 10 feet. The minimum angle of depression shall be 25 degrees from the horizontal datum. The minimum angle of elevation shall be 20 degrees from horizontal datum. Telescopes not equipped with a clutch shall be equipped with a locking mechanism in the upright position.

3.7.1 Cap. A removable, protective cap for the objective end of the telescope and an aperture cap for the eyepiece shall be furnished. The eyepiece cap shall have a dark finish to prevent disturbing reflections.

3.7.2 Prismatic eyepiece. A prismatic eyepiece giving an erect but reversed image shall be furnished.

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3.7.3 Diameter of exit pupil. The diameter of the exit pupil shall be not less than 2 millimeters (mm).

3.8 Reticle. The reticle and supporting parts shall be such that distortion will not be introduced when collimation is being adjusted. Capstan headscrews shall be used for adjusting the position of the reticle. The reticle hairs shall be not less than 0.00013 inch nor more than 0.00020 inch in width. The hairs shall not exhibit noticeable unevenness and shall appear opaque. Unless otherwise specified (see 6.2), the hairs shall be the contractor's standard type. Unless otherwise specified (see 6.2), the reticle pattern shall conform to figure 1. Stadia lines shall provide the horizontal distance from the vertical axis of the instrument to the rod within an accuracy of plus or minus 0.3 percent of all distances more than 100 feet. The ratio of the fixed stadia hairs shall be 1 to 100.

3.9 Circular level. A circular level shall be mounted either on the base plate or the table head forward of the pedestal, for approximating the level of the planetable. A ring shall be inscribed on the cover glass. This ring shall circumscribe the bubble at all temperatures. The diameter of the glass shall be between 0.50 and 0.75 inch. The sensitivity shall be 10 to 20 minutes per 2 mm when tested as specified in 4.5.2.9.

3.10 Alidade blade. The alidade blade or straightedge shall be brass conforming to ASTM B 36, UNS C21000 H02, or aluminum conforming to ASTM B 209, UNS A96061-O-T6. The top surface of the blade shall be finished as specified in 3.14. The underside of the blade shall be hard anodized. A base plate knob shall be located at each end of the blade. The fiducial edge of the blade shall be beveled and parallel to the axis of the telescope.

3.11 Compass. The compass shall be the induction damped trough type and shall be located at the eyepiece end of the telescope. It shall have a needle not less than 2.75 inches long. The compass shall be fitted with a dustproof, removable, nonmagnetic cover containing a window through which the needle is visible. The compass box shall be fitted with a needle release system. The needle shall have a single index mark at its south end which will match with a single index mark on the interior of the box when the compass is properly leveled, balanced, and centered. A means shall be provided for balancing the needle. The compass shall be fitted with a high grade hardened pivot and jewel bearing. The interior of the compass box shall have a flat white finish. The finish of the exterior shall be the same as that of the alidade. The compass shall have a mean error not more than 1/2 degree when tested as specified in 4.5.2.10.

3.12 Carrying case. The alidade shall be provided with a carrying case of the contractor's standard type, designed and constructed to protect the alidade from damage by handling incident to storage and transportation. Compartments shall be provided within the case for holding accessories so that they will not come in contact with each other or the alidade. The case shall be equipped with a hinged cover and shall be provided with a secure fastening device.

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3.13 Identification marking for DoD only. The alidade shall be identified in accordance with MIL-STD-130.

3.14 Treatment and painting. The alidade and case shall be treated and painted in accordance with the contractor's standard practice.

3.15 Type III, optical-reading, self-leveling, high standard.

3.15.1 Telescope. The magnification shall be between 14 and 24 diameters. The telescope shall have a minimum effective objective aperture of not more than 36 mm.

3.15.2 Reticle. The reticle shall be as specified in 3.8 except that the reticle pattern shall be on glass and shall conform to figure 1. The lines thereon shall be sharp, true, and clear when viewed through the eyepiece. Both the horizontal and the main stadia lines shall be spaced and the reticle shall be positioned so as to subtend a length of the rod equal to 0.01 of the horizontal distance from the center of the alidade to the rod, with a tolerance of plus or minus 0.3 percent. A cover glass shall be cemented over the graduated reticle. The reticle pattern shall have four red triangles positioned opposite the ends of the horizontal centerlines to facilitate quick and positive recognition of the centerlines.

3.15.3 Self-leveling mechanism. The vertical arc or its index shall be controlled by a self-leveling mechanism constructed to withstand commercial shipping and handling. The sensitivity of the self-leveling mechanism shall be such that the vertical arc can be reset to a given graduation and the telescope resighted at a fixed rod 200 feet distant, with deviations from the mean rod reading averaging not more than 0.01 foot (10 seconds). The self-leveling mechanism shall function accurately when the planetable is inclined from the horizontal not more than 1/2 degree in any direction. Provision shall be made for adjustment of the vertical circle or its index to the horizontal. To permit quick and accurate readings, the self-leveling mechanism shall be damped so that primary pendulum oscillation is stopped in not more than 5 seconds of time.

3.15.4 Vertical-angle graduations. The vertical-angle graduations shall be on glass, graduated as follows:

(a) Degree scale: The degree scale shall be graduated and numbered from 60 to 120 degrees to read 60 degrees when the telescope is inclined upward and 90 degrees when at horizontal. Each degree shall be numbered.

(b) Vertical Beaman graduations (V-scale): Forty divisions shall be graduated each way from the horizontal. Every even graduation shall be numbered, with 50 at horizontal in exact agreement with the 90 degree graduation line. The numbering shall increase with inclination of the telescope upward. These graduations shall be adjacent to the degree graduations. The number 50 on the V-scale shall be accentuated in bolder style.

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(c) Horizontal Beaman graduations for stadia distance correction: Twenty divisions shall be graduated and numbered each way from the horizontal, with the numbers from 80 to 100 to 80, with 100 at horizontal.

3.15.5 Reading microscope. The reading microscope shall have a magnification of not less than 22 power. A reticle mask with three windows shall separate the graduations in the field of view of the microscope. The microscope reticle degree scale shall be subdivided in 2-minute intervals. The image of the graduations shall be sharp and clear.

3.15.6 Alidade blade. The length of the blade shall be between 11.5 and 18 inches and the width shall be between 3 and 3.375 inches. Graduations shall be as specified (see 6.2).

3.15.7 Carrying case. The carrying case shall have an attached carrying strap of the contractor's standard type.

3.15.8 Weight and height. The weight of the alidade shall be not more than 8 pounds. The height shall be 8.25 inches, plus or minus 1 inch. The pedestal shall be not less than 3.5 nor more than 5 inches high.

3.16 Type IV, optical-reading, self-leveling, low standard.

3.16.1 Telescope. The telescope shall be as specified in 3.7. The magnification shall be between 10 and 24 diameters. The telescope shall have a minimum effective objective aperture of not more than 35 mm.

3.16.2 Reticle. The reticle shall be as specified in 3.15.2.

3.16.3 Self-leveling mechanism. The self-leveling mechanism shall be as specified in 3.15.3.

3.16.4 Vertical-angle graduations. The vertical-angle graduations shall be as specified in 3.15.4.

3.16.5 Reading microscope. The reading microscope shall be as specified in 3.15.5.

3.16.6 Alidade blade. The blade dimensions shall be 3.5 inches wide by 11.5 inches long. One edge of blade shall be graduated 50 parts to the inch, the other edge shall be graduated 4 inches to the mile.

3.16.7 Carrying case. The carrying case shall have an attached carrying strap of the contractor's standard type.

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3.16.8 Weight and height. The weight of the alidade shall be not more than 8 pounds. The height shall be between 2.5 inches and 4.5 inches.

3.17 Workmanship. Fabrication shall assure proper functioning of all parts of the alidade.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article alidades shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article alidades shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4. Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.1.1 Inspection lot. All units of the same type, offered to the Government at one time, shall be considered a lot for the purpose of inspection.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

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4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2 through 4.5.2.11.2, as applicable. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The alidades shall be examined as specified herein for the following defects:

Major:

101. Material not as specified.
102. Optical system and mounting not as specified.
103. Telescope not as specified.
104. Reticle not as specified.
105. Circular level not as specified.
106. Compass not as specified.
107. Carrying case not as specified.
108. Workmanship not as specified.

Minor:

201. Identification marking missing, incomplete, or illegible.
202. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Environmental.

4.5.2.1.1 Low-temperature. Subject the alidade to a temperature of minus 25° F for a period of time required for the largest internal mass in the alidade to attain temperature stability. All mechanical movements of the alidade shall be tested. Any evidence of damage or any motion not free shall constitute failure of this test. Lower the temperature to minus 28° F and maintain at this temperature for a period of not less than 6 hours. Allow the temperature to return to normal and examine the alidade. Any evidence of damage shall constitute failure of this test.

4.5.2.1.2 High-temperature. Subject the alidade to a temperature of 110° F for a period of time required for the largest internal mass in the alidade to attain temperature stability. Operate all motions to both limits and examine the instrument for malfunction. Evidence of any motion not free to both limits or any malfunction shall constitute failure of this test. Raise the temperature to 145° F and maintain for not less than 4 hours. Allow the temperature to return to normal and examine the alidade. Any evidence of damage or inoperation shall constitute failure of this test.

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4.5.2.1.3 Solar radiation. Temperature stabilize the alidade at 110° F and apply a source of directed radiant energy simulating solar radiation of 355 Btu/sq ft/hr to the instrument. A suitable source for this energy is five 300-watt R-40 reflector-spot incandescent lamps (or equal) mounted on 6-inch centers in the form of a cross with one lamp in the center common to the limbs. Such a source, with the plane of the cross horizontal, mounted directly over the alidade at a height of 38 inches from the center lamp to the top of the vernier control level will give a close approximation of radiant flux. After not less than 4 hours exposure, examine the alidade. Any evidence of damage or inoperation shall constitute failure of this test.

4.5.2.1.4 Humidity. Subject the alidade to one complete humidity cycle in accordance with MIL-STD-170. After the cycle has been completed, examine the instrument. Any evidence of damage or inoperation shall constitute failure of this test.

4.5.2.2 Beaman scale. Prepare a test chart consisting of 0.03-inch-wide lines ruled at 14.29, 20.41, 33.33 and 50.00 inches from a base line of the same width. The lines may either be ruled both above and below the base line, or the chart may be inverted during the test and the alidade being tested releveled on the base line. Set up the test chart perpendicular to the optical axis of the alidade and 100 inches from the center of the alidade being tested. After being leveled on the base line, sight the alidade according to the following schedule:

Vertical angle		Beaman readings	
Degrees	Minutes	Horizontal	Vertical
56	34 1/	80	90
48	26	90	80
41	32	96	70
38	08	98	64.1
30	00	100	50
21	52	98	35.9
18	28	96	30
11	36	90	20
3	26	80	10

1/ To be omitted where the range of observation is less than 27 degrees from the horizontal.

Any sighting of the horizontal falling outside of one of the ruled lines at any setting shall constitute failure of this test. Any other test giving equivalent accuracy may be substituted for this test.

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4.5.2.3 Stadia. Test all stadia intervals, by reading on a calibrated Philadelphia level rod at not less than three distances differing by not less than 100 feet. The minimum distance shall be not more than 100 feet, and the maximum distance shall be such that the rod can be read clearly with the telescope focused at infinity. Any reading intercept differing by more than 0.3 percent from the actual distance shall constitute failure of this test.

4.5.2.4 Magnification. Measure the magnification power of the telescope in diameters as follows: Focus the telescope for parallel light with aid of a low-power auxiliary telescope placed behind the ocular of the telescope under test. Without change of focus, place a transparent graduated scale in front of and as near as practicable to the objective lens. Replace the low-power telescope with a measuring microscope and focus on the image of the scale. The magnifying power of the telescope is the ratio of a given length of the scale to the length of the image in the exit pupil. A magnification not within the limits specified in section 3 for the applicable type shall constitute failure of this test.

4.5.2.5 Minimum diameter of exit pupil. With the telescope adjusted as specified in 4.5.2.2, run the microscope, mounted on the micrometer, in and out until focused on the smallest sharply defined bright circle visible through the microscope. Measure the diameter of this circle which is the diameter of the exit pupil. As an alternate method, mount a screen behind the telescope eyepiece and project parallel light through the telescope. An appropriate light source and collimator may be used to obtain parallel (collimated) light. Move the screen back and forth until the smallest sharply defined circle of light is obtained on the screen. Carefully measure the diameter of this circle to obtain the diameter of the exit pupil. Any exit pupil measuring less than 2 mm shall constitute failure of this test.

4.5.2.6 Field of view. With prismatic eyepiece attached, focus the telescope on a vertical rod or graduated scale which extends through the entire field of view along the vertical crosshair. Read the limiting rod or scale values just within the field of view. Set the horizontal crosshair first on the lower and then on the upper limit of the field of view, reading the vertical angle to each setting on the vernier scale of the vertical arc. A difference of less than 1 degree 30 minutes between the two readings shall constitute failure of this test.

4.5.2.7 Minimum focus. Set the telescope at minimum focus and move a graduated scale along the line of sight until the markings are sharply defined. Measure the distance from the scale along the optical axis to approximately the centerline of the pedestal or standard. Any dimension greater than 10 feet shall constitute failure of this test.

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4.5.2.8 Resolution. Observe Pattern No. 28 of NBS Special Publication 374 set 100 feet from the objective of the telescope. The limiting angle of resolution is greater than 6 seconds if the pattern cannot be clearly resolved. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.9 Sensitivity of levels. Test the circular level on a level trier or hinged beam, one end of which is moved vertically by means of a micrometer screw, each revolution of which results in a known angular movement of the beam. Accompany each trial by a 180-degree reversal trial. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.10 Compass needle sensitivity. Place the alidade on a stable, level surface and orient it on magnetic north by centering the compass needle carefully on the zero mark. Set a target on the line of sight at a known distance of not less than 100 feet. Pivot the alidade about any selected point on the fiducial edge of a blade and again carefully center the compass needle. Set another target in the new line of sight at the same distance as the first target. Measure the linear distance between the two targets to determine the angular error. (A distance of 0.873 foot between two targets 100 feet from the instrument subtends an angle of 1/2 degree.) A mean error more than 1/2 degree shall constitute failure of this test.

4.5.2.11 Self-leveling mechanism tests for type III and type IV alidades.

4.5.2.11.1 Sensitivity. Sight the instrument on a graduated rod fixed at a position 200 feet from the instrument. At a given graduation on the instrument, read the rod. Make at least four resettings and resightings in which the planetable is inclined not more than 1/2 degree in different directions. Readings deviating from the mean by more than 0.01 foot shall constitute failure of this test.

4.5.2.11.2 Damping. With the telescope clamped at any random vertical inclination, set the alidade on a stable, level surface. When reading the vertical arc in the normal manner, any discernible primary oscillation of the pendulum of more than 5 seconds of time after the alidade has come to rest on the level surface shall constitute failure of this test.

4.6 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be Level A, or Level B, or Commercial, as specified (see 6.2).

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5.1.1 Level A. Each completed alidade shall be preserved in accordance with the level A preservation and packaging requirements of MIL-S-3534.

5.1.2 Level B. Each complete alidade shall be preserved in accordance with the level B preservation and packaging requirements of MIL-S-3534.

5.1.3 Commercial. Each complete alidade shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A or level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Alidades, preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-S-3534.

5.2.2 Level B. Alidades, preserved as specified in 5.1, shall be packed in accordance with the level B packing requirements of MIL-S-3534.

5.2.3 Commercial. Alidades, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking (see 6.2).

5.3.1 Military agencies. Marking shall be in accordance with MIL-STD-129.

5.3.2 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.3 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Surveying alidades are used for stadia surveying. This includes measuring distances, directions and determining elevations. Self-leveling alidades (type III and type IV) are for use where speed of operation and accuracy are of prime importance.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) When a first article is required for inspection and approval, and number of units required (see 3.2).
(d) Type of reticle hairs required if other than specified (see 3.8).
(e) Type of reticle pattern required if other than specified (see 3.8).
(f) Specific blade graduations required (see 3.15.6).
(g) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(h) Type marking required (see 5.3).

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first produced alidade. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each alidade.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of alidades.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.7 Classification change. The type I and type II alidades have been discontinued since they are no longer required.

MILITARY INTERESTS:

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY: Army - ME

Project 6675-0715

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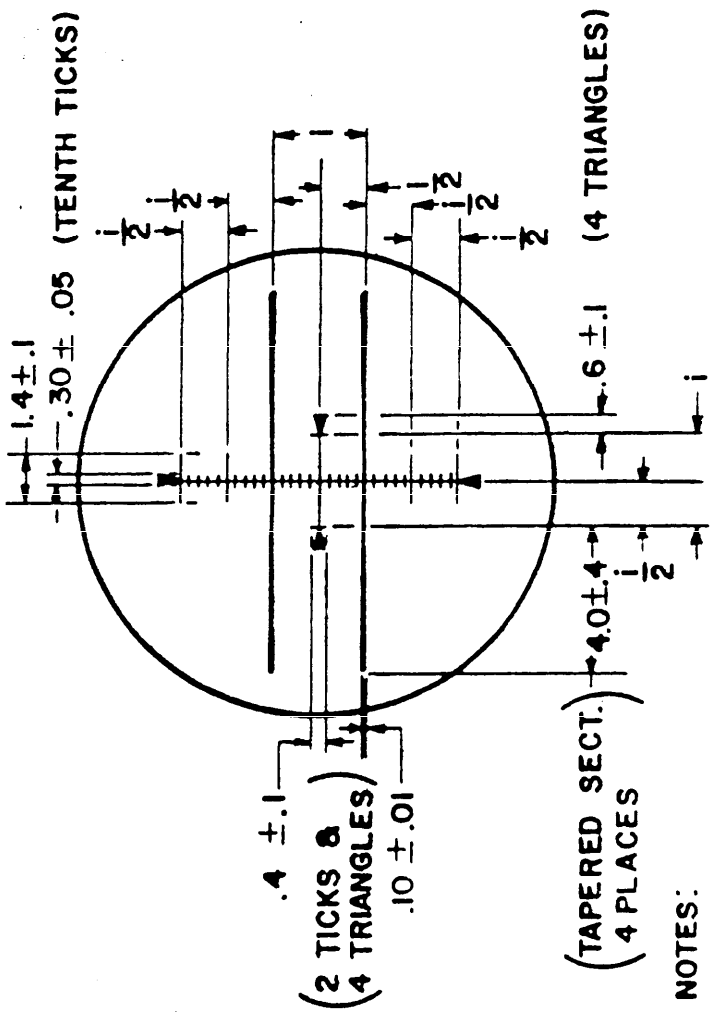


FIGURE 1. Reticle for alidades, Types III and IV.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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